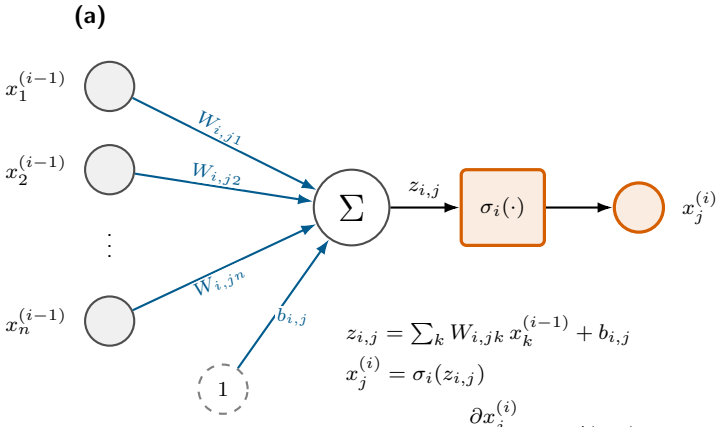
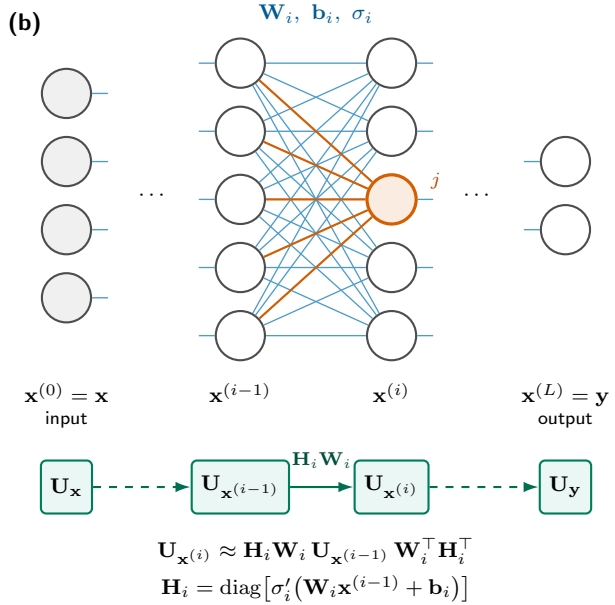

(a)
$x_1^{(i-1)}$
$x_2^{(i-1)}$
$x_n^{(i-1)}$
$W_{i,j1}$
$W_{i,j2}$
$W_{i,jn}$
$b_{i,j}$
1
$\sum$
$z_{i,j}$
$\sigma_i(\cdot)$
$x_j^{(i)}$
$z_{i,j} = \sum_k W_{i,jk}\, x_k^{(i-1)} + b_{i,j}$
$x_j^{(i)} = \sigma_i(z_{i,j})$
sensitivity: $\frac{\partial x_j^{(i)}}{\partial x_k^{(i-1)}} = \sigma_i'(z_{i,j})\, W_{i,jk}$


(b)
$\mathbf{W}_i,\ \mathbf{b}_i,\ \sigma_i$
j
$\mathbf{x}^{(0)} = \mathbf{x}$
input
$\mathbf{x}^{(i-1)}$
$\mathbf{x}^{(i)}$
$\mathbf{x}^{(L)} = \mathbf{y}$
output
$\mathbf{U}_\mathbf{x}$
$\mathbf{U}_{\mathbf{x}^{(i-1)}}$
$\mathbf{H}_i\mathbf{W}_i$
$\mathbf{U}_{\mathbf{x}^{(i)}}$
$\mathbf{U}_\mathbf{y}$
$\mathbf{U}_{\mathbf{x}^{(i)}} \approx \mathbf{H}_i\mathbf{W}_i\, \mathbf{U}_{\mathbf{x}^{(i-1)}}\, \mathbf{W}_i^\top \mathbf{H}_i^\top$
$\mathbf{H}_i = \mathrm{diag}\left[\sigma_i'\left(\mathbf{W}_i\mathbf{x}^{(i-1)} + \mathbf{b}_i\right)\right]$